National University of Computer and Emerging Sciences, Lahore Campus

Course Software for Mobile Devices Course
Program: BSCS
Semester: Spring 2026
Assignment 3

Important Instructions:

1. "Submit your `app/src` folder compressed as a `.zip` file" named as your roll number., i.e., 22L-1111.zip
2. **You are not allowed to copy solutions from other students. We will check your code for plagiarism using plagiarism checkers. If any sort of cheating is found, heavy penalties will be given to all students involved.**
3. Late submission of your solution is not allowed. After the deadline, no submission will be accepted.

Assignment 3 – FastMart E-commerce site

Important Instruction

1. You can hardcode product images. But there should be at least 6-8 different product images.
2. For Bonus marks you can store image urls on firebase realtime db and use Glide to load images.

Github

You are required to create a GitHub repository and regularly push your project throughout development. The project must follow a proper MVVM architecture, with clear separation of concerns and each functionality organized into appropriate folders. Your code should be well structured and include meaningful comments explaining important logic. A detailed README file must be included, describing the project, its features, setup instructions, and overall architecture; however, this README must be written by you without using AI-generated content, as this will result in marks deduction. Additionally, you are expected to maintain a clean and meaningful commit history with multiple commits reflecting your development progress repositories with very few commits or a single commit will be penalized.

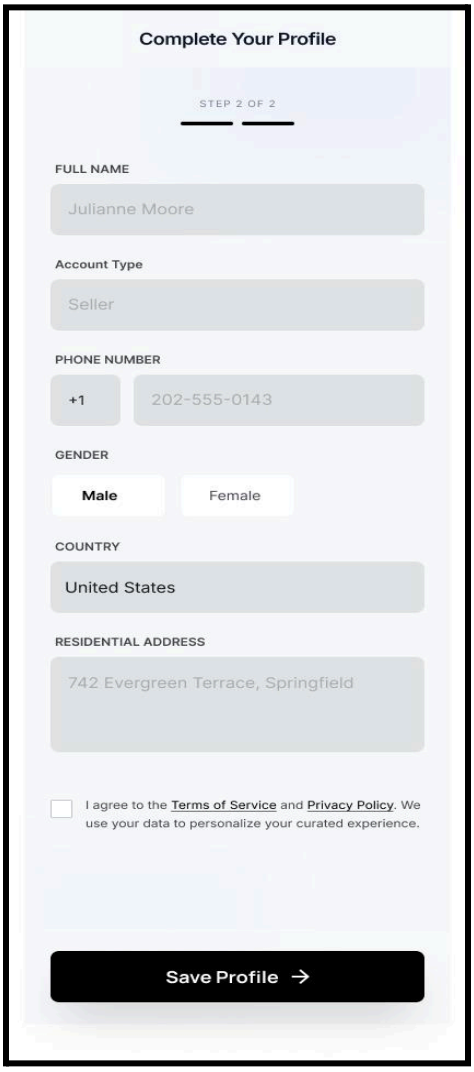
Figma Link:

https://www.figma.com/design/twPZcW5tTxTE97IYpF0L1o/SMD-Assignment-3?node-id=0-1&t=1V2mlAUu6ZN5lkLg-1

Instructions

1) All the colors and strings should be stored in colors.xml and strings.xml. Hard coded colors would result in marks reduction.
2) All UI elements (Buttons, TextViews, RecyclerViews, etc.) must have meaningful and consistent IDs.
3) All functions and variables should have meaningful names.
4) Proper indentation, spacing, and commenting. Key functions and complex logic must have comments explaining purpose and workflow.

Login/SignUp

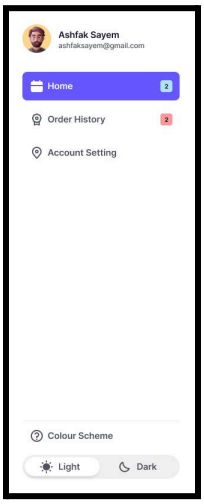


During the SignUp, the user is asked for name, address, gender, date of birth, phone number, country and **accountType** (Seller or Buyer) in addition to email and password. You can keep the

previous UI structure. Now change the SharedPref based Login/SignUp to Firebase Authentication based Login/SignUp. For SignUp use **createUserWithEmailAndPassword** create user and then store the user info (name, address, phone no, country, address, gender, accountType) in Firebase RealTime Database. During Log In use **signInWithEmailAndPassword** for log in and then get the user info from Firebase Realtime db to check the accountType and depending on that show the respective (buyer, seller) account. Use sharedprefs to store user id, name, isLoggedIn and account so next time user do not need to log in rather user is automatically logged.

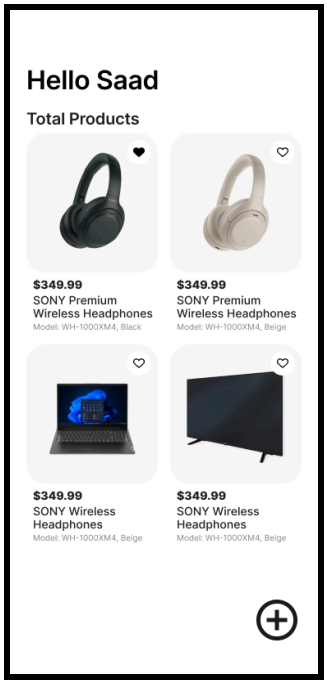
For sellers

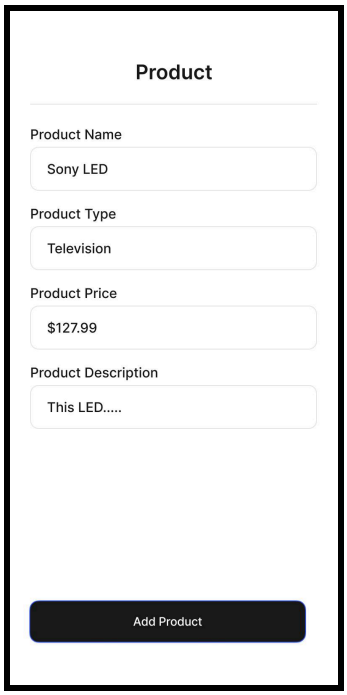
Drawer Navigation



1. Just like how there is bottom navigation in the buyer's account, implement the drawer navigation in the seller one which has a home, order history page and accounts page.
2. On pressing a light or dark theme the colors are changed across the app. Also store the theme in shared pref and default theme is light.

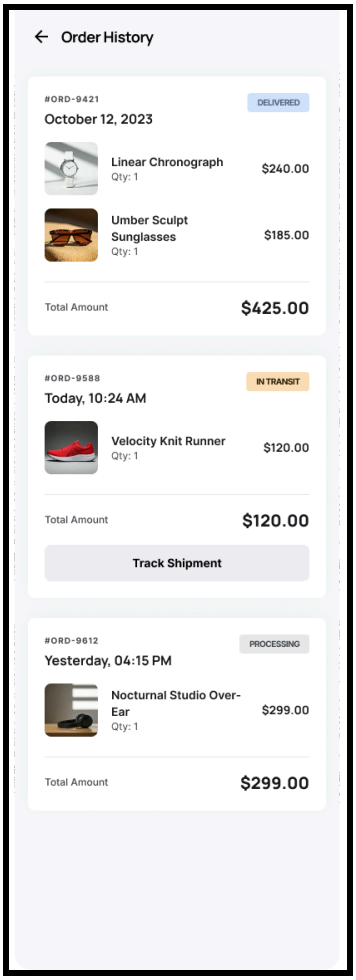
Seller Home Page





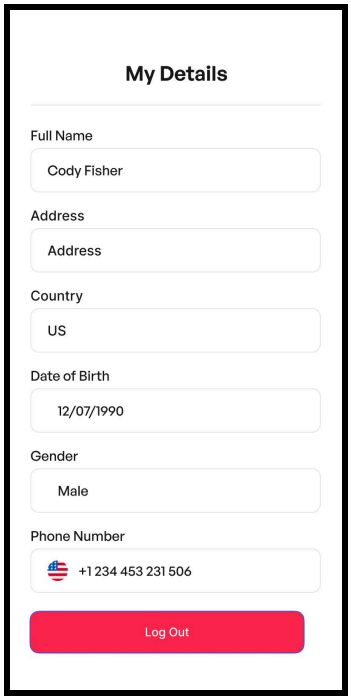
1. There is a floating button which when pressed moves the seller to the ProductAdd screen where the user can add the product and it will be stored on firebase realtime db.
2. On the home screen all already added products are shown.
3. Sellers can click the product to see the ProductDescription page.

Order History Page



1. When a buyer checks out the order is stored in order history with name of products, quantity etc. (See cart screen in this assignment)
2. This information is displayed in the order history page.
3. Use a recycler view to show orders. Each order contains the name of products ordered with quantity and cost of each and total cost.
4. You DO NOT need to use another RecyclerView inside each order item.

Account Page



The name, address, gender, date of birth, phone number, country which was stored in firebase realtime db during the sign up is to be loaded and displayed here for the specific user logged in. There is a button for logout when clicked the user is logged out from both firebase auth and shared prefs and navigated to sign in screen. Also user info is also removed from firebase real-time db.

For buyers

New Home Page

1) All product data must be fetched dynamically using Firebase Realtime Database.
2) The application should implement real-time data synchronization such that:
    a) Newly added products by the seller are immediately reflected on the Buyer's Home Screen.
    b) Buyers should not need to restart the application to see updated products.
3) Use Firebase listeners (e.g. ValueEventListener) to continuously listen for changes in the database and update the UI automatically whenever new products are added, modified, or removed.
4) Ensure proper data structure in Firebase

Favourites

1. All favourite products must be stored locally using **SQLite**.
2. When a user clicks the heart icon on any product, that product should be added to the Favourites database.

3. Any changes to favourites must be reflected immediately: When a product is added it appears instantly in the Favourites page. When removed it disappears immediately without requiring app restart.
4. When the user clicks the triple-dot icon, it shows an AlertDialog "Do you want to delete this product from favourites?". On "Yes" remove products from SQLite and on "No" do nothing.
5. When the **cart icon** is pressed. The product should be added to the Cart with quantity=1.

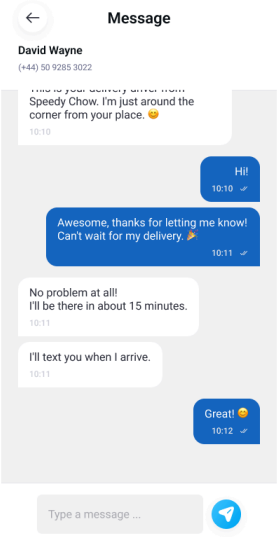
Cart

1. All cart products must also be stored locally using **SQLite** with quantity.
2. Products can be added to the Cart from:
    a. Product Description Page (Buy Now button)
    b. Favourites Page (cart icon)
3. Each cart item should include "+"/"-" buttons to increase/decrease quantities. Update the corresponding record using Update query.
4. Total price calculation should fetch the latest data from SQLite. Calculate total price and Update total price TextView instantly.
5. Each item should have a three dot option which on clicking deletes the item from SQlite database using Delete query.
6. On clicking the Checkout button, all cart items must be fetched from **SQLite** and an order summary should be prepared including product names, quantities, individual prices, and the total price. This summary is then sent via SMS to the specified number.
7. After successful checkout, the order details must also be stored in the Firebase Realtime database as part of the user's order history. Only the seller can view order history.

Account

The name, address, gender, date of birth, phone number, country which was stored in firebase realtime db during the sign up is to be loaded and displayed here for the specific user logged in. Like if a user is logged in the name, address, gender, date of birth, phone number, country is displayed.
There is a button for logout when clicked the user is logged out from both firebase auth and shared prefs and navigated to sign in screen. Also user info is also removed from firebase real-time db.

Chat Functionality



There should be a floating button on the home page which when pressed shows a chat with seller option. For chat with sellers, you can use Firebase RealTime db to implement real-time messaging between buyers and sellers where each message includes sender ID, receiver ID, message text, and timestamp. Chat history must be stored and displayed(You can and cannot use RecyclerView as there is an easier way to deal with this UI), with a clear distinction between sender and receiver messages.